

Battle of the Bits			
	Blue Laser	Holographic Storage	Millipede
Storage capacity	30 GB per layer per side	Terabyte class	80 GB per square cm
Technology used	400 nm wavelength laser	Optics and holography	Nanotechnology
Data transfer speed	Medium	Very high	High
Cost of implementation	Relatively low	High	Medium
Time to implement	1-2 years	5-7 years	4-6 years
Applications	Enhanced DVD applications	Data mining and warehousing	Storage for mobile and portable devices
Companies involved	Hitachi, LG Electronics, Matsushita, Pioneer, Philips, Samsung, Sharp, Sony and Thomson Multimedia	IBM, Rockwell, Lucent	IBM

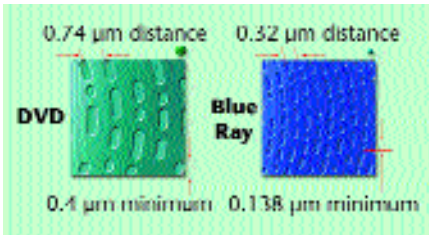
DVD-ROM drives have a wavelength between 630 nm and 650 nm (1 nm = 1 millionth of a millimetre), which puts them in the red band of light. The reason why blue lasers are so exciting is because their wavelength lies in the 400 to 450 nm range. This shorter wavelength light translates into disks written with blue laser being capable of storing between 25 to 30 GB of data per layer per side! That gives it five to six times the storage density of existing DVDs. Imagine having all the *Star Wars* movies with full-blown surround audio and crystal-clear video quality on a single disk with room to spare!

If it is simply a question of using laser light with a smaller wavelength, you might ask why this wasn't done before. The reason for this is that the materials used to generate blue lasers have a relatively shorter lifespan compared to those used for red lasers. While blue lasers are still in the research phase, there are three types of methods that are used to generate these lasers:

Zinc Selenide (ZnSe): The initial method for implementing blue lasers involved the use of Zinc Selenide to fabricate the diodes that generate blue lasers. However, this material has a relatively short lifespan and its power requirements make it economically unsuitable for commercial implementation. Also, these lasers have wavelengths ranging from 460 to 520 nm, putting them at the end of the

blue and closer to the green light band of the spectrum.

Gallium Nitride (GaN): This material has proved to be very successful in the creation of blue lasers and has generated wavelengths as low as 370 nm with relatively high reliability. Most of the work in blue lasers today is based on this material.



The shorter wavelength of the blue laser allows for more data than on a DVD

Second Harmonic Generation lasers: These lasers are relatively new on the blue laser scene but have exhibited very high levels of reliability. Through this intelligent method, the frequency of a given laser is doubled (that is, the wavelength is halved) and laser light within the blue spectrum is generated. This is done through an apparatus called a Distributed Bragg Reflector (DBR) where, for example, the frequency of an infrared laser with a wavelength of 850 nm is doubled, resulting in a blue laser with a wavelength of 425 nm.

It will be some time before blue laser technology becomes commercially viable. The major hindrance to this technology is the cost of implementation. A Blue-ray device available today would cost about Rs 2 lakh! The second hurdle is that of reliability: red lasers that are used in all CD-ROM and DVD-ROM drives today have a life cycle of about 10,000 hours. Now compare that to the meagre hundreds of hours that the Gallium Nitride-based blue lasers last for using today's technology! However, it is just a matter of time before these issues are addressed. Scientists predict that in just a couple of years nearly all new optical storage devices would be based on blue lasers.

Looking ahead

While the very thought of the storage capacity offered by these path-breaking technologies is overwhelming, there will soon come a time when we will think nothing of being able to copy a 10 GB computer game onto a terabyte-class storage device. But this is going to take time. All these technologies are still only limited to the confines of a handful of well-guarded research laboratories. These technologies will be available for mass consumption only when the materials and the technologies used to fabricate these mammoth storage systems become economically viable. ■

MARCO D'SOUZA

1997

Seagate introduces the world's first fibre channel interface disc drive

1998

IBM introduces the first hard drive with giant magnetoresistive (GMR) heads

IBM launches the Microdrive

2000

Blue Laser storage is demonstrated

Seagate introduces world's first 15,000 rpm disc drive

Seagate reaches a landmark capacity with the 180 GB Barracuda hard drive

2001

First use of a hard disk drive in a game console- the Xbox

2002

Seagate demonstrates world record areal density - 100 qigabit per square inch

IBM debuts Millipede for the next generation of data storage